HDOC Day-64

NAME-PRIYADARSHNA

SAPID-590032928

BATCH-15

Aim

To write a C program to find the only repeated element in a variable-length integer array using pointers, functions, and the XOR technique.

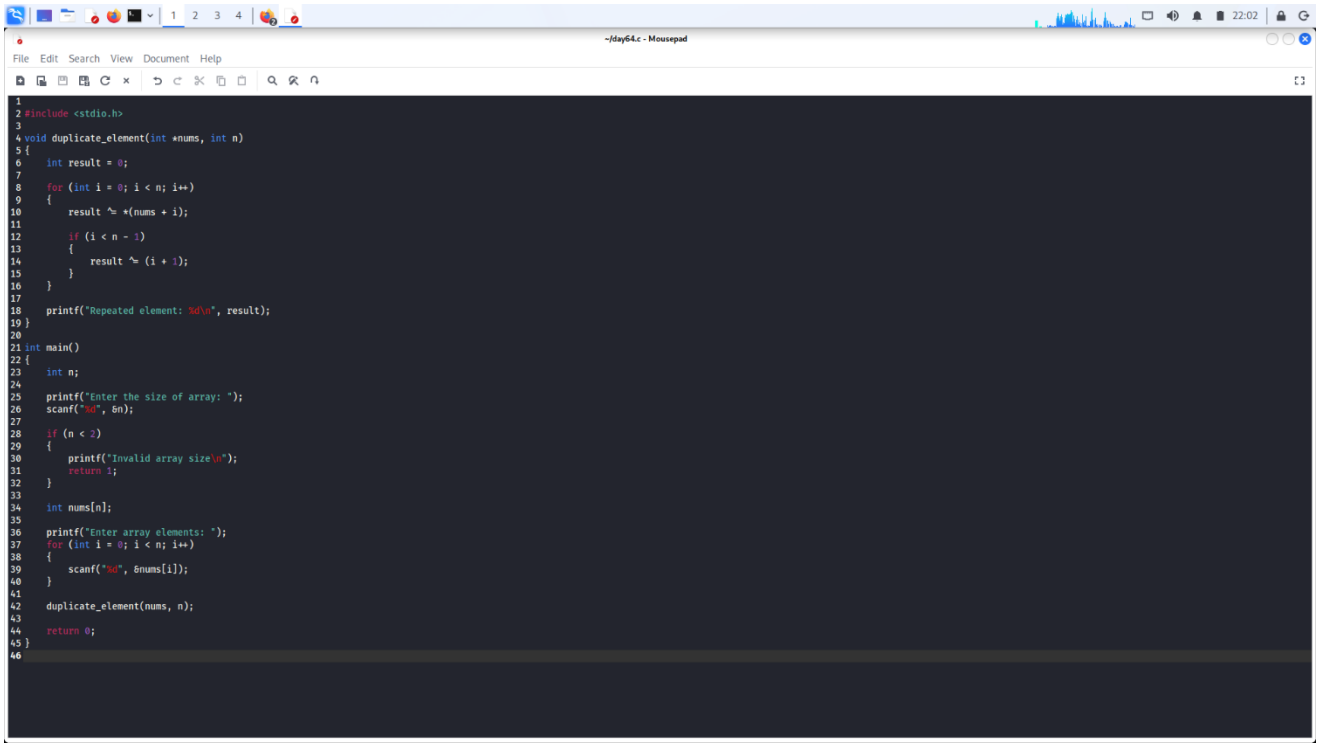
Question

Write a C program that reads the size and elements of a variable-length integer array nums[]. The array contains exactly one repeated element. Pass the array to a function named duplicate_element() using pointers.

Algorithm

1. Start the program.
2. Read the size n of the array.
3. Declare a variable-length array nums[n].
4. Read all the array elements.
5. Pass the array address and size to duplicate_element().
6. Initialize result = 0.
7. Use a loop to XOR all array elements.
8. In the same loop, XOR the numbers from 1 to n-1 (the expected distinct values).
9. The final XOR result gives the repeated element.
10. Display the repeated element.
11. Stop the program.

>C PROGRAM:



>OUTPUT:

